ASSOCIATED DETAIL SPECIFICATION

RESISTORS, FIXED, FILM, PRECISION, ESTABLISHED RELIABILITY, STYLE RFP05

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for established reliability, precision, film, fixed resistors. Resistors are available within the full range of reactance, tolerance, and failure rates.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-R-122 - Resistors, Fixed, Precision, Established Reliability, General Specification for.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Electronics Supply Center, ATTN: DESC-ES, 1507 Wilmington Pike, Dayton, OH 45444-5276 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5905

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3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with MIL-R-122, and as specified herein.

3.2 Design and construction. Resistors shall be of the design, construction, and physical dimensions specified on figure 1.

3.3 Power rating. The power rating shall be .05 watt at 85°C (see 6.2). The applicable power rating curve is curve C.

3.4 Voltage rating. The continuous working voltage shall not exceed 200 volts.

3.5 Resistance. Minimum and maximum resistance values shall be as follows:

a. Minimum resistance: 10 ohms.

b. Maximum resistance: .25 megohm.

* 3.6 Weight. The maximum weight shall not exceed .001111 pound (.5 gram).

3.7 Overload. Apply 6.25 times rated power to the resistor for 5 seconds. When tested as specified, the applied voltage shall not exceed 500 volts.

3.8 Dielectric withstanding voltage. The applicable test voltages are as follows:

a. Atmospheric pressure: 450 volts.

b. Barometric pressure (reduced): 200 volts.

3.9 Current noise. The current noise index shall not exceed -20 dB maximum.

3.10 Thermal EMF. The thermal EMF shall not exceed 5.0 microvolts per degree Celsius.

* 3.11 Marking. Due to size limitations, this style resistor shall be marked with the following information, as a minimum:

042AM: Date code, reactance, resistance tolerance and failure rate.
4475A75J: Resistance value with temperature characteristic and JAN marking.

The complete marking is required on the unit package. Where manufacturers are able to provide more information, the following is preferred in the sequence presented:

Military part number, style, manufacturer's trademark, production lot code, and source code.

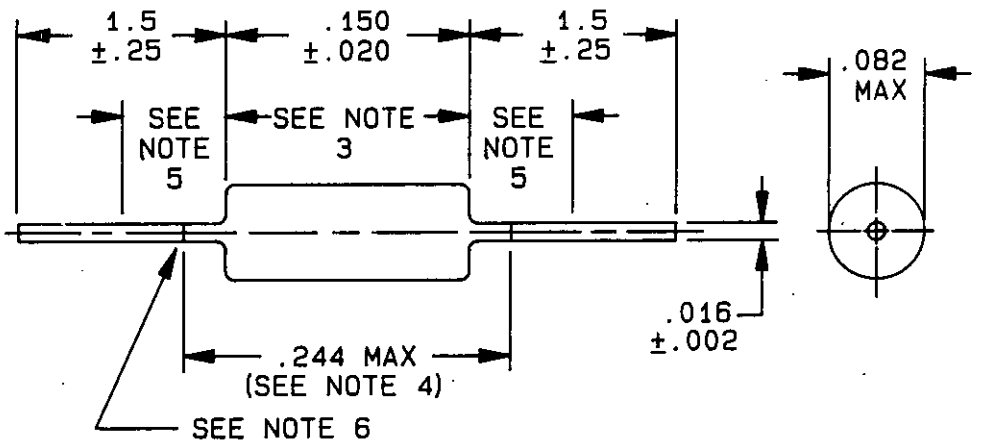
4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-R-122.

5. PACKAGING

5.1 Packaging. Packaging requirements shall be in accordance with MIL-R-122.

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Inches	mm
.002	0.05
.016	0.41
.020	0.51
.082	2.08
.150	3.81
.244	6.20
.25	6.4
1.5	38.0

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The end of the body shall be that point at which the eyelet diameter equals the nearest drill size larger than 250 percent of the nominal lead diameter.
4. Maximum length is "clean lead to clean lead".
5. Resistance measuring point shall be .5 ±.125 inch (13 ±3.18 mm) for resistance values of 10Ω or more and .625 ±.025 inch (15.88 ±0.64 mm) for resistance values less than 10Ω.
* 6. Lead concentric tolerance is to be measured at the point of lead egress from the resistor body. Lead to be concentric to the resistor body within .016 (0.41 mm) TIR.

FIGURE 1. Fixed resistor, precision.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-R-122 are applicable to this specification.

6.2 Power rating at 25°C. The power rating at 25°C for 2,000 hours of life test has been established at .1 watt, 200 volts, maximum voltage. It should be noted that the failure rate level is established at the single condition of 85°C, .05 watt and 10,000 hours life test duration.

6.3 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Navy - AS, OS
Air Force - 17, 85, 99
DLA - ES

User activities:
Army - AT, AV, ME
Navy - CG, MC
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5905-1166)

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